

CS470 2026 Spring Midterm Prep Note

Please review all "question slides" as well.

If you have limited time, you may want to review the following points first.

(WARNING!) However, midterm exam questions will not be limited to the points below.

Midterm exam structure

1. T/F Questions 1~2 Sets
2. Problem sets (algorithms, short answers, discussions): 2~4 sets
3. (new) Open/discussion questions: several subproblems or at most 1 set.

1. Introduction

- Superalignment (discussion)

2. Basic concepts in AI/ML

- (recap) Superalignment (discussion)
- **Turing test**
- An Extremely Brief History of AI
- **AGI, Weak AI vs Strong AI**
- (Q) What's our strategy to prove our value in the world of Machines,
- (Q) Can we find/prove that there is a task/domain that humans outperforms any advanced AIs?

3. Uninformed Search

- Planning ~ search: DFS, BFS, UCS
- Define "reflex agent"? rational? (world IS) <> Planning Agents? (world WOULD BE, what if?), consequence of actions
- **(Q) Search problem (simplified world)**
 - State space
 - actions and costs
 - Successor function
 - start state
 - goal test
 - solution: a sequence of actions
- **(Q) State Space Graphs vs Search Tree**
- **(Q) DFS vs BFS: Optimality, Time complexity, space complexity**
- **Uniform Cost Search: Optimality, Time complexity, space complexity**
- **(Q)** Between DFS and BFS, which one is a complete algorithm ? assuming the action cost is uniform.

4. Uninformed Search

- UCS
- Priority queue
- (Q) What's the difference between iterative deepening and DFS?
- (Q) Among BFS, DFS, and UCS, which one finds the optimal solution regarding total cost?
- (Q) What if minimum arc cost is negative? Does UCS work?

5. Informed Search

- **Heuristics:** Admissibility, Dominance, **Relaxed problem**
- Greedy Search
- **A***
 - Termination
 - Optimality: Proof!
 - **Admissible Heuristic: Relaxed Problems**
- **Semi-Lattice of Heuristics**
 - "Trivial" admissible heuristic
 - $h = h''$? **max** of multiple admissible heuristics?
- Graph Search
 - **Consistency of Heuristic**
 - PF: Consistency $\rightarrow$ Admissibility
- (Q) What is an optimal **best(?)** heuristic?

6. Game Trees

- Adversarial Games, Zero-sum Games, General Games
- **Minimax Search**
 - **Depth-Limited Search**
 - **Alpha-Beta Pruning**
 - **Evaluation Function**
- (Q) Does Alpha-beta pruning change the decision?
- (Q) Does cooperation emerge from Min-max trees? How can we extend the alpha-beta pruning to multi-agent game?

7. Uncertainty and Utilities

- **Expectimax Search**
- Utilities & Transformation
 - minimax is invariant to monotonic transformations
 - (Q) expectimax is NOT invariant to **monotonic transformations**
- **Utilities / Multi-Agent Utilities**
- **Rational Preferences**
- (Q) Can we prune an expectimax search tree?
- (Q) Given one episode, when ghosts respond randomly, does

- expectimax yield optimal moves?
- (Q) Does the discrepancy between assumptions and reality yield suboptimal strategies?

8. MDPs -I

- definition of Markov Decision Processes,
- Discounting: Discounted utility
- Stationary preferences
- Infinite utilities -> finite utilities: Finite horizon, discounting, absorbing state
- **Value iteration**
- **V, Q functions**
- (Q) Can a policy generalize to unseen/never-visited states?
- (Q) MDPs assume the agent always knows its current state. How realistic is this assumption (e.g., robots, self-driving cars) and what's the cost of relaxing it?

9. MDPs -II

- **Bellman Equations**
- **Convergence proof**
- Policy Evaluation
- Policy Extraction
- **Policy iteration**: the policy often converges long before the values
 - **Policy evaluation**
 - **Policy improvement** ~ policy extraction
 - Comparison with Value iteration
- Online planning vs. Offline planning
 - Exploration
 - Exploitation
 - Regret
 - Sampling
- (Q) What is the complexity per iteration of policy evaluation? Is it more efficient than a value iteration?
- (Q) Write a linear system equation that addresses Policy Evaluation.